SIRC-1 EXCEPTION ENTRY AND RETURN

ENTRY

Commits only after the vector has been fetched

1 Select

At an instruction boundary, take the highest-priority eligible exception. For a retryable fault, the instruction was cancelled and its remaining phases ran as no-ops.

2 Fetch vector

Read the high word, then the low word, at vector × 2 in the vector table. If the fetch fails, a fault is raised instead and none of the later steps happen.

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steps 3–5 take effect together

3 Save state

Write the return address, SR and current level to the exception's link register (index = priority − 1 for registers 0–6). Retryable faults save the faulting instruction so RETE retries it; interrupts, traps and trace faults save the next instruction.

4 Update state

SR.P = 0 (supervisor mode), SR.T = 0 (no tracing), SR.EA = 1 (handler active). Current exception level = level of this exception.

5 Jump

Load the program counter with the vector target; the handler starts.

Handler runs

Runs in supervisor mode at the new exception level.
ETFR and ETTR can read or edit the saved state before returning.
A higher-priority exception can interrupt it, using its own link register.

RETURN (RETE)

1 Restore state

RETE restores p, sr and the exception level from the current link register. SR.EA is then set if the restored level is non-zero, and cleared if it is 0.

2 Resume

Execution continues at the return address.
A pending exception that is now eligible is serviced before the next instruction.